Constancy RoPE: An Empirical Study of Spectral Attention Gains at 4K

Working research report

October 17, 2026

Abstract

We test whether spectral feedback on rotary-position attention gains improves training-free 4K generation with FLUX.1-Krea-dev. Constancy RoPE compares the predicted clean latent spectrum with a native-resolution reference and applies a bounded, noise-gated correction to spatial attention gains. Two completed campaigns fail to demonstrate an improvement over the released SEGA baseline. In the second campaign, only SEGA plus time-matched constancy at gain $\eta = 1$ passes the eight-prompt development gate. On 187 held-out prompt pairs, its ImageReward difference is $+0.00857$ with a 95% confidence interval of $[-0.00841, 0.02742]$, while full-resolution MUSIQ changes by -0.20860 $[-0.70887, 0.31512]$. PickScore and crop MUSIQ decrease. These results do not support an efficacy or state-of-the-art claim. This working report records the completed experiments; it is not a submission-ready methods paper.

1 Question and tested controllers

The project asks whether feedback toward a model's native latent spectrum can improve resolution extrapolation through per-band attention gains. The target image size is 4096×4096, compared with a 1024×1024 native calibration size. The baseline is the released SEGA [2] allocation and RoPE base, identified in the implementation as `sega_official`. The released allocation differs from the paper-equation implementation; the two are not interchangeable baselines.

The first campaign tested the original constancy controller, then a time-matched native spectral target. The latter was also tested on SEGA's RoPE base with its reference temperature. This did not establish an improvement. The second campaign tested two effective-pool temperature rules, `alias_temp` and `alias_temp_sref`, and a composition of released SEGA with time-matched constancy, `sega_ctm`, at $\eta = 0.5$ and 1.0.

For the composed controller, let m_d be SEGA's released query/key multiplier for rotary pair d. A correction in spectral band b is

$$e_b = \tfrac{1}{2}\big(\log P_{\mathrm{native},b}(\sigma) + a - \log \widehat{P}_b\big), \tag{1}$$

$$\delta_b = \eta w_b \ \mathrm{clip}(e_b, -1, 1), \tag{2}$$

$$g_d = m_d^2 \exp\big(\mathrm{interp}(\delta)_d\big). \tag{3}$$

Here a is the anchor offset, w_b is the existing visibility gate, and $\widehat{P}_b$ is measured from the preceding transformer call's clean estimate. The native target is interpolated at the current scheduler noise level. Query and key are each multiplied by $m_d \exp(\delta_d/2)$. The first step has no correction, the index axis retains unit gain, and no additional global temperature is applied. Spectral powers use orthonormal FFTs and means within bands. The vision analogy motivates this controller; it does not establish its efficacy or novelty.

2 What the gain algebra establishes

Pair scaling and its causal limit. For fixed content vectors, the rotary-position logit [3] can be written as a finite trigonometric sum

$$\ell(\Delta) = \frac{1}{\sqrt{d_k}} \sum_j \mathrm{Re}\big[c_j e^{i\theta_j \Delta}\big]. \tag{4}$$

Applying a shared multiplier m_j to the query and key coordinates of pair j multiplies that term by m_j^2. Its sign and phase still depend on c_j. Increasing a high-frequency coefficient therefore does not by itself establish more local attention, higher image-band power, or sharper decoded detail. The three existing Krea development probes have inconsistent response signs: one prompt has no positive decoded-width diagonal among six tested bands. Those probes jointly perturb both spatial axes on NTK2, so they also cannot identify a single-axis response under released SEGA.

Standardized profile shape. Let ℓ be a nonconstant vector of sampled log powers and let $z(\ell) = (\ell - \operatorname{mean} \ell)/\operatorname{std}(\ell)$, using one fixed standard-deviation convention. For $\beta > 0$,

$$z(a\mathbf{1} + \beta\ell) = z(\ell). \tag{5}$$

This elementary identity applies to an ideal standardized-profile allocation. It does not imply invariance of SEGA's radial spread or its complete gain. The paper-inspired reference implementation adds a standard-deviation epsilon, and the released implementation floors energy and standard deviation. Their behavior is therefore invariant only under the corresponding inactive-floor or negligible-stabilizer conditions, up to floating-point error. The statement is about a formed log-power profile; raising a two-dimensional spectrum to a power need not commute with averaging it into that profile.

Allocation and scalar magnitude are coupled. Before clipping, write a centered SEGA multiplier as $m_j = m_{\mathrm{ref}}(1 + a_j)$ with $\operatorname{mean} a_j = 0$. Then

$$\operatorname{mean}(m_j^2) = m_{\mathrm{ref}}^2\big[1 + \operatorname{mean}(a_j^2)\big]. \tag{6}$$

Thus centering query/key multipliers does not preserve their mean squared magnitude, which is the mean coefficient gain. Clipping changes these moments again. The new RMS controls isolate this effect per spatial axis. They do not match the actual attention-logit distribution, its variance, or its entropy.

Forward-noise expectation and the gate. For the linear data–noise interpolation used to motivate rectified flow [1], assume independent data x_0 and unit white noise ε, with $x_\sigma = (1 - \sigma)x_0 + \sigma\varepsilon$. With an orthonormal transform, channel summation over C channels, and mean-per-band density on non-DC bands,

$$\mathbb{E}_\varepsilon P_b(x_\sigma) = (1 - \sigma)^2 P_b(x_0) + \sigma^2 C. \tag{7}$$

The cross term vanishes in expectation. This is not an exact decomposition of one noisy sample or one generated ODE trajectory. The controller's statistic $\rho_b = \max(P_b(x_\sigma)/(\sigma^2 C) - 1, 0)$ and gate $w_b = \rho_b/(1 + \rho_b)$ are design choices inspired by this identity. They do not estimate the uncertainty of the delayed clean prediction. Likewise, zero correction at an anchored target spectrum is an algebraic property of the rule, not a stability theorem for the full sampling process.

3 Executed protocol

All reported comparisons use the FLUX.1-Krea-dev transformer with FLUX.1-dev text encoders and VAE, 4096^2 output, 28 sampling steps, distilled guidance 4.5, and scheduler shift $\mu = 1.15$. The selected candidate and baseline share the executed schedule, seed convention, and SEGA RoPE base. Earlier FLUX.1-dev outputs exhibited a lattice artifact and are excluded from these results.

The 195 Aesthetic-4K evaluation captions are split into development rows 0–7 and held-out rows 8–194. Native calibration uses 16 separate Pick-a-Pic prompts. Development eligibility requires positive mean differences in both ImageReward and full-resolution MUSIQ, with CLIP difference at least -0.3. At most two variants may advance; no settings change after selection.

The second campaign reports exactly 187 matched seed-0 prompt pairs. Generation produced 188 images because row 7 was included to preserve task chunking; that row is excluded from all held-out tables. Confidence intervals use 2,000 paired bootstrap draws with bootstrap seed 0 and the 2.5 and 97.5 percentiles. Intervals are nominal per metric, without multiple-comparison adjustment. An interval crossing zero does not establish equivalence.

4 Results

4.1 Development selection

Table 1 records selection only. Both alias variants lose ImageReward. The $\eta = 0.5$ composition misses the MUSIQ gate by 0.007850. Only $\eta = 1$ advances; development scores are not generalization evidence.

Table 1: Development means on eight prompts, seed 0.

Variant	ImageReward	MUSIQ	CLIP	Advance
Released SEGA	1.423018	42.488060	30.320004	Reference
Alias temperature	1.281098	35.807356	29.366986	No
Alias, reference gain	1.277426	44.972845	30.238460	No
SEGA + constancy, $\eta = 0.5$	1.441334	42.480210	30.200250	No
SEGA + constancy, $\eta = 1$	1.448742	42.505928	30.342451	Yes

4.2 Held-out comparison

Table 2: Held-out seed-0 means and paired differences, candidate minus released SEGA, over 187 prompts. Higher is better except the last three rows.

Metric	SEGA	SEGA + constancy	Difference [95% CI]
ImageReward	1.18994	1.19852	$+0.00857$ $[-0.00841, +0.02742]$
PickScore	23.05255	22.97729	-0.07526 $[-0.10234, -0.04914]$
CLIP	29.13552	29.04160	-0.09392 $[-0.19651, -0.00368]$
MUSIQ, full image	45.22784	45.01924	-0.20860 $[-0.70887, +0.31512]$
MUSIQ, 1024 crops	58.32778	57.19612	-1.13166 $[-1.38754, -0.90219]$
CLIP-IQA, full image	0.60299	0.60593	$+0.00294$ $[-0.00018, +0.00609]$
CLIP-IQA, 1024 crops	0.58597	0.58846	$+0.00250$ $[-0.00070, +0.00563]$
Spectrum log RMSE	1.06838	1.06596	-0.00242 $[-0.00905, +0.00433]$
Spectrum slope error	0.19995	0.19741	-0.00254 $[-0.00678, +0.00192]$
Tiling at 1024	0.06836	0.06844	$+0.00008$ $[-0.00072, +0.00081]$

The selected composition has no demonstrated gain on ImageReward or full-resolution MUSIQ. PickScore, crop MUSIQ, and CLIP have negative nominal confidence intervals. The remaining intervals cross zero. This second campaign therefore does not repair the original controller's failure.

The earlier time-matched controller on SEGA's base also regresses on the same 187 seed-0 prompts. Full-resolution MUSIQ changes by -3.03602, with interval $[-4.28169, -1.81488]$. Crop MUSIQ changes by -2.19266, with interval $[-2.81241, -1.56666]$. Its ImageReward difference is unresolved at $+0.01782$ $[-0.02271, 0.05913]$. A temperature-only control on SEGA's base reduces full-resolution MUSIQ by 2.08550 points, with a paired interval $[-3.24526, -0.96407]$. These comparisons constrain the tested designs; they do not show that all spectral controllers must fail.

5 Limits and continuation

The second-campaign candidate has one held-out seed and one evaluated backbone at one resolution. No second-seed replication of this candidate, new backbone, human preference study, or broader resolution claim is supported. Aggregate FID/KID scores from the raw 188-image evaluation include development row 7 and are excluded from the held-out conclusion. No new throughput comparison is claimed in this report.

The earlier extended theory and literature sections remain in the source tree but are excluded because they describe superseded settings and unresolved claims. Section 2 retains only qualified algebraic observations. The third diagnostic below tests gain magnitude, frequency allocation, and controller timing under a frozen development-only protocol. The already inspected held-out set must not become a tuning set presented as fresh evaluation.

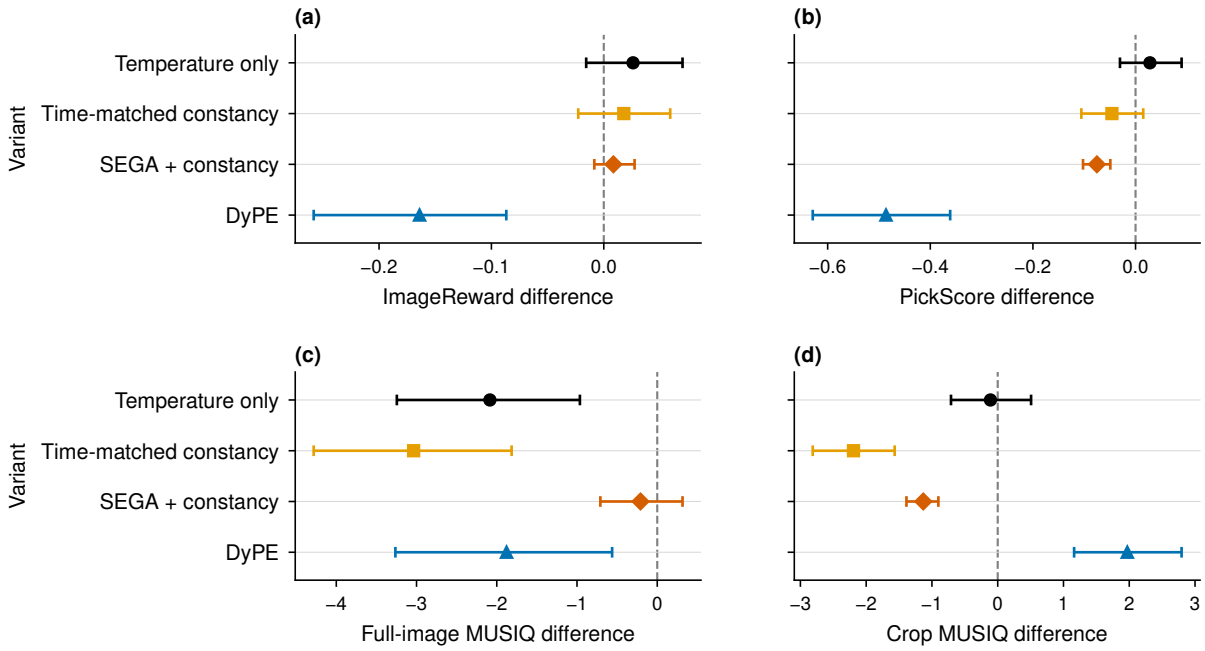


Figure 1: Neither constancy variant demonstrates a joint preference and quality gain over released SEGA. Points show paired differences; whiskers are nominal 95% prompt-bootstrap intervals from 2,000 draws over 187 prompts, single seed. Zero is the released SEGA baseline; higher is better in all panels. DyPE's crop quality gain accompanies a preference loss. The temperature-only and time-matched constancy variants use SEGA's RoPE base and reference temperature.

Prespecified development diagnostics, not results. A third diagnostic compares three controls on development rows 0–7: released-SEGA allocation normalized to the reference query/key RMS; spatially uniform gains matched to the released allocation's realized RMS; and the $\eta = 1$ constancy composition with its target evaluated at the delayed estimate's own scheduler noise level. A fresh released-SEGA run is the reference. These distinguish gain magnitude from frequency allocation and isolate target timing. The code passes 256 CPU tests, but no GPU outcome for these controls is available in this snapshot. The planned job contains 32 images and was queued on October 17. A positive development result would require separate confirmation.

Conditional confirmation protocol. Before the third development outcome, a separate set of 64 captions was frozen from Aesthetic-4K's 2048-source split after excluding normalized matches to all prior evaluation and calibration captions. Fixed hash ordering selected the captions without inspecting generation or scores. Any eligible candidate would use its unchanged settings at 4096^2 with a fresh SEGA baseline. The captions are fresh to this project's evaluation, with no model-training disjointness or semantic-near-duplicate guarantee. They do not supply matched 4K reference images. The prespecified analysis uses 10,000 paired prompt-bootstrap draws, seed 0, and common resampling indices across metrics and candidates. Nominal 95% intervals accompany all metrics. For the two primary metrics, ImageReward and full-image MUSIQ, Bonferroni-adjusted percentile intervals have coverage $1 - 0.05/(2K)$ for K selected candidates. Joint primary support requires both adjusted lower bounds to exceed zero. These are approximate bootstrap intervals, not exact finite-sample guarantees. Secondary losses remain part of the report; confirmation cannot select further controller settings. No confirmation result is available in this snapshot.

Evidence provenance. The completed campaign is recorded at implementation commit `a1ede1a` on October 16, 2026. Its archived paired table, per-prompt scores, and analysis protocol are in `.agent/notes/main2/`. The current held-out table is generated from those CSVs after checking its means and paired intervals against the per-prompt records. This check reuses saved scorer outputs; it is not a fresh image-scoring reproduction. The previous session records 194 passing CPU tests. No new inference or training result is included in this snapshot.

References

[1] Xingchao Liu, Chengyue Gong, and Qiang Liu. Flow straight and fast: Learning to generate and transfer data with rectified flow. In *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2209.03003.

[2] Javad Rajabi, Kimia Shaban, Koorosh Roohi, David B. Lindell, and Babak Taati. SEGA: Spectral-energy guided attention for resolution extrapolation in diffusion transformers, 2026. URL https://arxiv.org/abs/2605.22668. arXiv:2605.22668.

[3] Jianlin Su, Yu Lu, Shengfeng Pan, Ahmed Murtadha, Bo Wen, and Yunfeng Liu. RoFormer: Enhanced transformer with rotary position embedding. *Neurocomputing*, 568, 2024. arXiv:2104.09864.